UBER

Machine Learning Case Study

Integration • Applications • Impact • Challenges

PRANAUV RAAJ N

1. WHY UBER NEEDED MACHINE LEARNING

Uber is a real-time marketplace where traffic, rider demand and driver availability change continuously.

Traditional rules and historical averages cannot capture every road and marketplace condition.

ML learns patterns from trip, location, traffic and behavioural data.

Objective: make faster, more accurate predictions and improve the rider/driver experience.

REAL-TIME DATA

- GPS
- Traffic
- Time
- Driver supply
- Demand
- Trip history
- Routes

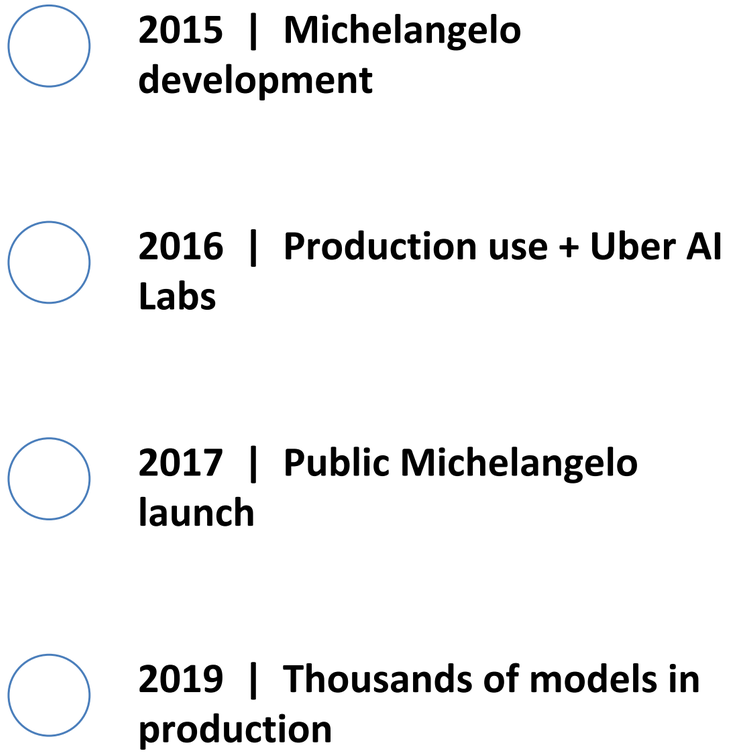
2. WHEN DID ML INTEGRATION START?

2015 — Uber began building Michelangelo in mid-2015 as its ML needs grew.

2016 — Michelangelo was already serving production use cases; Uber also acquired Geometric Intelligence and created Uber AI Labs.

2017 — Uber publicly introduced Michelangelo as its internal ML-as-a-service platform.

2019 — Uber described Michelangelo as supporting thousands of production models and a broad range of ML use cases.



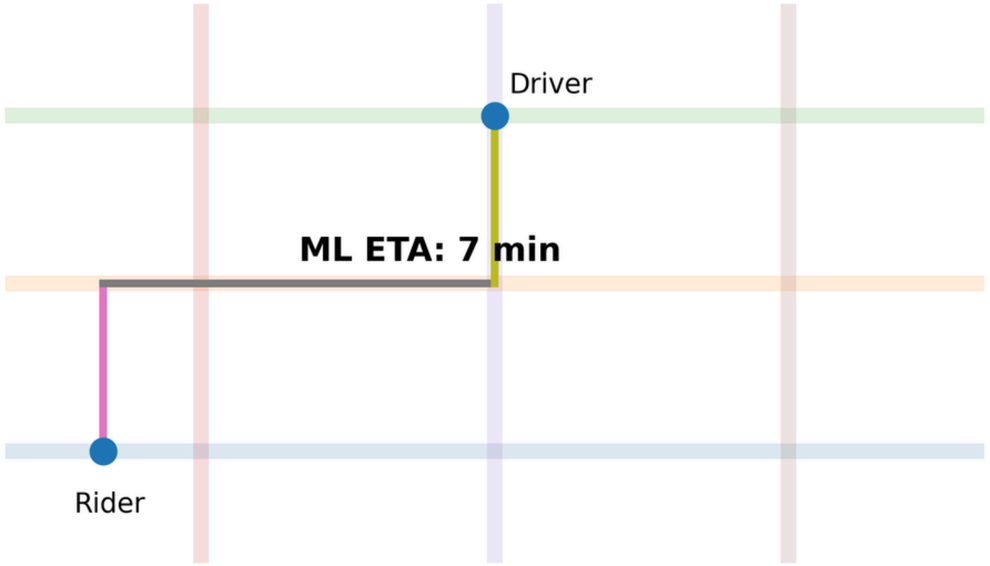
3. HOW UBER USES MACHINE LEARNING

ETA prediction — estimates driver arrival and trip duration.
Demand forecasting — predicts where and when rides will be requested.

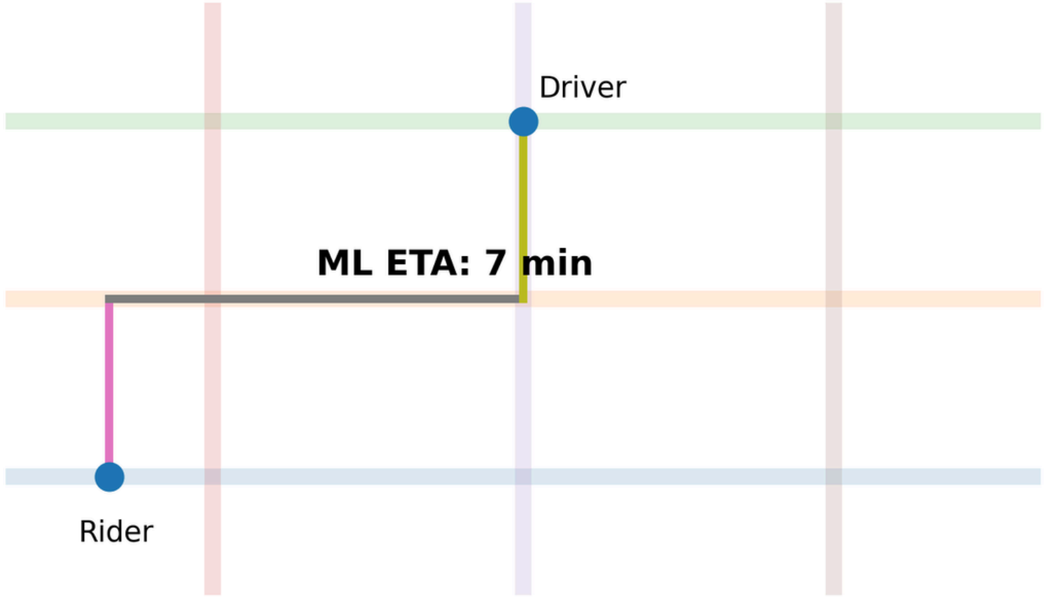
Matching — helps connect riders with suitable available drivers.

Pricing & routing — supports marketplace and route-related decisions.

Fraud, safety and Uber Eats — identifies suspicious patterns, predicts delivery times and supports ranking.



4. ETA & DEMAND FORECASTING — REAL-WORLD EXAMPLE

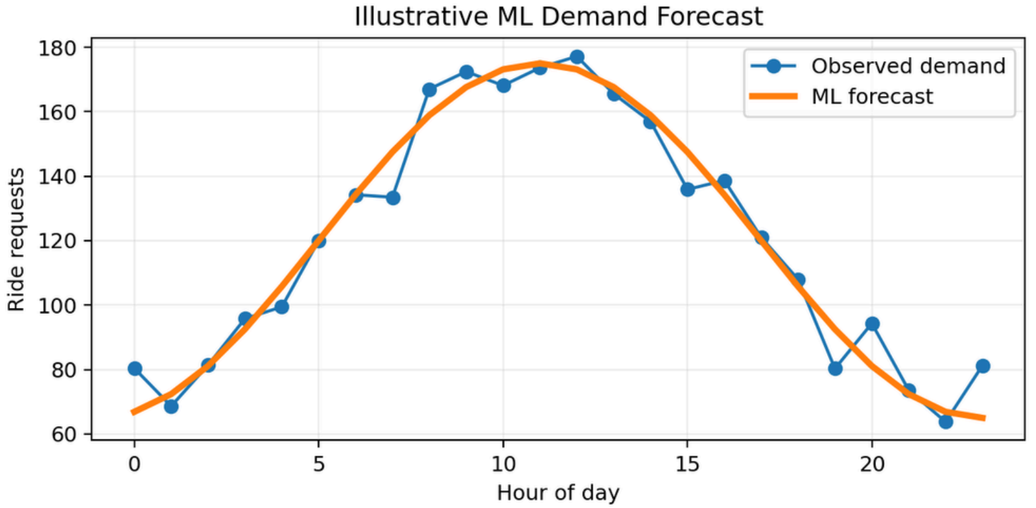


ETA models use signals such as route, traffic, location, time and historical patterns.

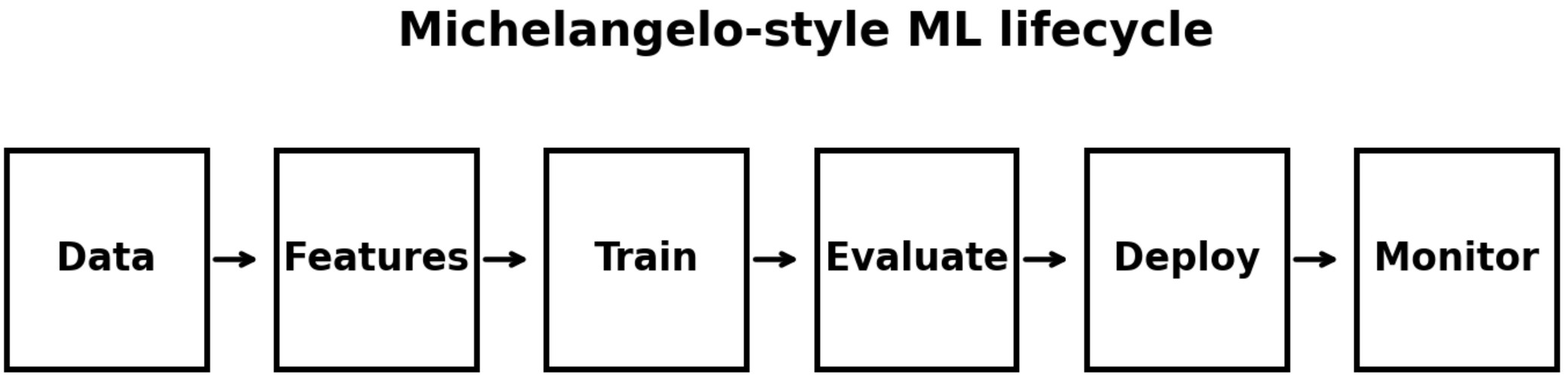
Uber reported ML-based ETA correction reducing average ETA error by more than 50% in some cases.

Demand forecasting models help anticipate supply-demand imbalance.

Uber reported 2%–18% improvement over previous algorithms in a cited LSTM demand-forecasting application.



5. MICHELANGELO — UBER'S ML PLATFORM



1. Standardises the end-to-end ML lifecycle.
2. Supports model training, evaluation, deployment, online serving and monitoring.
3. Helps teams move from experimentation to reliable production systems.
4. Uber's platform evolved to support classical ML, time-series forecasting and deep learning.

6. IMPACTS & CHALLENGES

IMPACTS

- More accurate ETAs
- Better rider experience
- Improved demand planning
- Faster, standardised ML development
- ML embedded across rides, delivery and marketplace

CHALLENGES

- Real-time latency
- Huge data volumes
- Changing traffic and demand
- Data quality & feature consistency
- Deployment and monitoring complexity
- Fairness / transparency in algorithmic decisions

7. CONCLUSION & KEY TAKEAWAYS

Uber demonstrates how ML can move from prediction to real-time business decision-making.

ML supports ETA, demand forecasting, matching, routing, pricing, fraud and delivery use cases.

Michelangelo was a major step toward scaling ML across teams and production systems.

Key lesson: successful ML requires accurate models PLUS strong data, deployment, monitoring and engineering infrastructure.

KEY SOURCES

- Uber Engineering — Michelangelo
- Uber Engineering — Demand & ETR Forecasting
- Uber Engineering — ML model representation
- Uber Help — How ETAs work